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ENGINEERING



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AI-Powered Vulnerability Detection and Secure Code Review Platform

**CSA5802 - DEVSECOPS ENGINEERING AND APPLICATION
SECURITY**

TEAM MEMBERS

Krithika M (192521385)

Prathiksha SN (192565090)

Shreya V (192565018)

Sri Janani B(192421036)

INTRODUCTION

- Modern software applications are increasingly vulnerable to cyber threats due to insecure coding practices and rapidly evolving attack techniques.
- Manual security reviews are time-consuming, error-prone, and difficult to perform consistently for large software projects.
- DevDeployShield is an AI-powered platform designed to automate vulnerability detection and secure code review before deployment.
- The platform enables developers to upload project files, analyze source code, identify security weaknesses, and generate intelligent recommendations.

PROBLEM STATEMENT

- Many developers deploy applications without performing comprehensive security analysis.
- Existing manual code review processes require significant time and experienced security professionals.
- Security vulnerabilities such as hardcoded credentials, SQL Injection risks, weak configurations, and outdated dependencies often remain undetected.
- Traditional vulnerability assessment tools are complex and may not provide developer-friendly recommendations.
- There is a need for an intelligent automated platform that can quickly detect vulnerabilities and provide actionable security improvements before deployment.



ENGINEER TO EXCEL

OBJECTIVE



- To develop an AI-powered secure code review platform that automatically detects vulnerabilities and improves application security before deployment.
- Upload and validate project files securely before analysis.
- Automatically scan source code and identify security vulnerabilities using rule-based and AI-assisted techniques.
- Analyze detected vulnerabilities, calculate project security score, and classify project risk levels.
- Generate detailed security reports and provide recommendations to help developers fix identified issues.



Existing System vs Proposed System

Existing System

Manual code review process

Time-consuming security analysis

Requires security experts

Limited vulnerability identification

Minimal security guidance

Proposed System

Automated AI-powered vulnerability scanning

Fast and intelligent project scanning

Easy for developers to use

Automatically detects multiple security vulnerabilities

Provides security recommendations, risk level, and security score

Module 1- Project Upload and Secure File Management Description

This module provides a secure interface for uploading Java or Spring Boot projects in ZIP format. It validates the uploaded file, stores it safely on the server, extracts the project contents, and prepares the project for further security analysis.

Functions

ZIP project

validationUploa

d statusZIP

extractionProject

storageProject

initialization

Input Java/Spring Boot ZIP Project

Output Extracted project ready for scanning.

Module 2 AI-Powered Vulnerability Detection Engine Description

- This module performs static source code analysis by scanning every file inside the uploaded project. It detects security vulnerabilities using predefined security rules and pattern matching techniques.

Functions

Recursive file scanning

Pattern matching

Security rule

Vulnerability identification

Severity classification

Output List of detected vulnerabilities.

How It Works

Source Code Upload



AI Scans & Analyzes the Code



Detects Security Vulnerabilities



Generates Report with Risk Level & Fix Suggestions

Module 3 Secure Code Review and Risk Assessment Description

- This module evaluates the vulnerabilities detected during scanning and determines the overall security posture of the uploaded project.

Functions

Count vulnerabilities

Calculate security score

Risk level determination

Project statistics

Security recommendations

AI recommendation generation

Output Security Score

Work flow

Source Code Review



Identify Security Risks



Assign Risk Level (Low/Medium/High/Critical)



Provide Secure Coding Recommendations

Module 4 Security Dashboard and Report Generation Description

- This module presents the scan results in a user-friendly dashboard and generates downloadable security reports for developers.

Functions

Interactive dashboard

Vulnerability table

Security score

visualization

Risk visualization

PDF report generation

Output

- Security Dashboard
- PDF Report
- Scan Summary

Tools & Technologies Used

- Java 21 – Core programming language for backend implementation.
- Spring Boot – Web framework for building the DevDeployShield application.
- Thymeleaf + HTML + CSS + JavaScript – Frontend development and user interface.
- Maven – Dependency management and project build automation
- ZIP Utilities & Java File I/O – Project upload, extraction, and source code scanning.



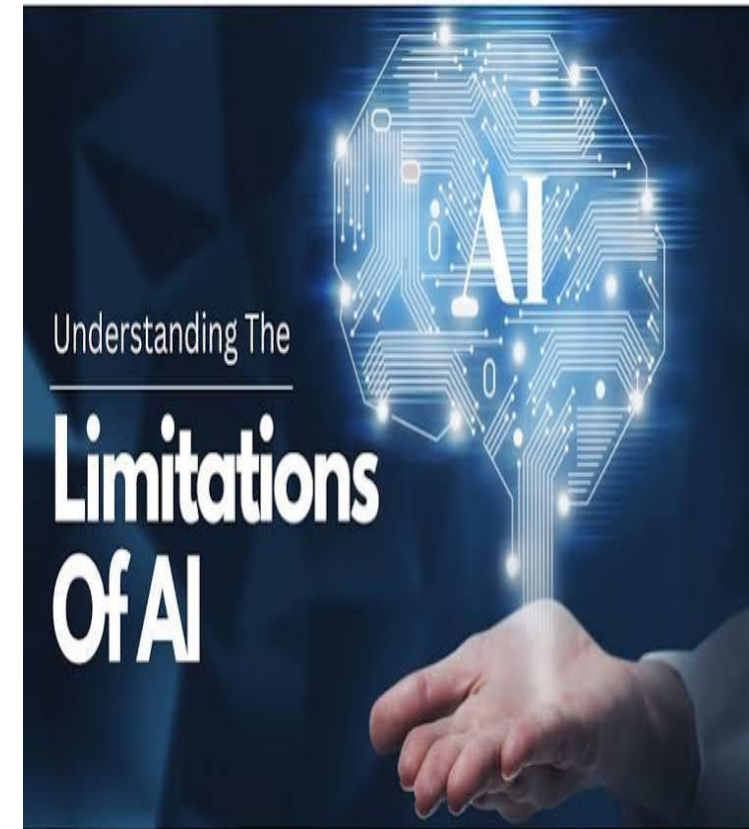
ADVANTAGES

- Automates vulnerability detection and secure code review
- Reduces manual effort and improves development efficiency.
- Provides developer-friendly security recommendations.
- Improves application security before deployment.
- Supports DevSecOps practices by integrating security into development.



LIMITATIONS

- Currently supports only Java and Spring Boot projects.
- Uses rule-based detection for many vulnerabilities in current version.
- Does not perform dynamic runtime security testing.
- Large projects may require additional scanning time.
- Advanced AI models require more computational resources for deeper analysis.



CONCLUSION

- DevDeployShield simplifies secure software development by automating vulnerability detection.
 - The platform improves application security through intelligent code analysis and secure coding recommendations.
 - Developers can identify and fix security issues before software deployment.
- The project demonstrates practical implementation concepts using Spring Boot.
- Future enhancements will include machine learning–based vulnerability prediction, real-time monitoring, cloud deployment, and support for multiple programming languages.

**THANK
YOU**

